



CreditBridge

PSB Hackathon 2026 | UCO Bank x Ministry of Finance

AI-Powered Scoring

Alternate Credit Scoring System

AI-Powered Credit for the Invisible

Unlock fair loans for 190M+ borrowers with no formal credit history using consent-based alternative data and multi-agent AI scoring.



Abhishek Kumar
Team Leader



Ankur Gupta
Team Member

Consent-first

AES-256 Encrypted

Explainable AI

DPDP Act 2023

SCORE RANGE

300-850

↑ Like CIBIL scale

PROCESSING TIME

60s

⚡ Fast approval



KEY VALUE PROPOSITIONS



6 Data Signals
Alternative scoring



190M+
Potential
borrowers

— The Invisible Borrower Crisis

190 Million Indians Are Invisible to Banks

A reliable street vendor earning ₹25,000/month is denied a ₹1 lakh loan because they have no formal credit history. The system cannot help people who were never part of it.



Ravi Kumar's Story

Tea Stall Owner, Guwahati

Ravi has run his tea stall for 5 years, earning ₹25,000/month consistently. He pays his phone bill on time every month, buys supplies from the same merchants, and has never defaulted. Yet when he asked UCO Bank for a ₹1 lakh loan to expand, they said **NO** — because he has no CIBIL score.

 5 years of business

 ₹25,000/month

 Perfect payment history

AFFECTED POPULATION

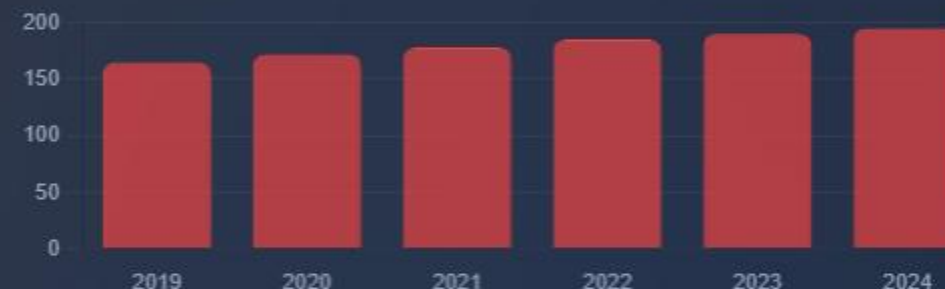
190M+

 1 in 7 Indians

CURRENT IMPACT

40-60%

 Moneylender rates



THE CIRCULAR TRAP

Need Credit
to build history



Need History
to get credit



Bank Rejection
No CIBIL = No loan



Predatory Rates
40-60% interest

The Opportunity

Inclusion + Profitable Lending

Unlock a new lending segment with 190M+ potential borrowers seeking ₹10,000-₹3,00,000 loans. Banks gain a profitable market while borrowers access fair credit.

Market Size & Timing

Addressable market with strong growth potential



190M+
Borrowers



**₹10K-
₹3L**
Ticket size



UPI Adoption

95%

↑ +12% YoY growth



Telecom Penetration

89%

✓ Ready for scoring



Bank Upside

New lending segment with lower CAC and controlled risk through better signals



Borrower Impact

Fair bank credit vs 40-60% moneylenders, build formal history, fuel MSME growth

CreditBridge's Wedge



60-second score



Multi-agent AI



Explainable



Privacy-first

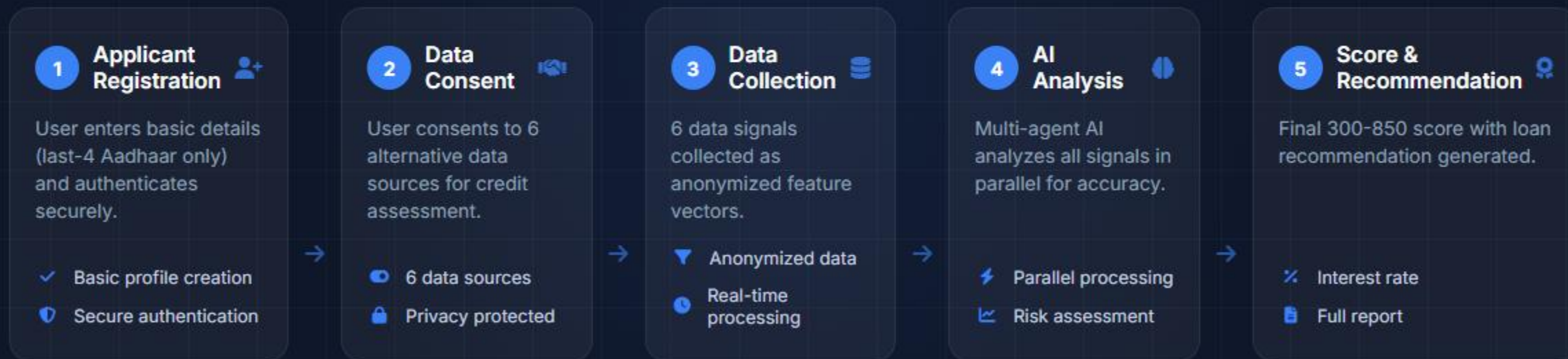
MARKET GROWTH: POTENTIAL BORROWERS (M)



Solution Workflow

From Consent to Credit in 60 Seconds

A seamless 5-step process that transforms applicant data into a reliable credit score with loan recommendation.



Phone Bill



E-commerce



Geolocation



Psychometric



Merchant



Cashflow



Credit Scoring Intelligence

The 6 Data Signals That Predict Creditworthiness

Each signal is scored 0-100 by a specialized AI agent, with weights optimized for accuracy and fairness to provide a comprehensive financial profile.



Phone Bill Payment

24+ months history

25%

Measures on-time/late payments, disconnections, and payment consistency over time.

✓ **Why it predicts:** Mirrors EMI discipline and financial reliability



Bank Cashflow

Regular credits

20%

Analyzes average balance, regular credits, bounced transactions, and savings behavior.

✓ **Why it predicts:** Income stability + spending discipline



Psychometric Assessment

20-30 questions

20%

Evaluates financial planning, debt attitude, integrity, and stress response patterns.

✓ **Why it predicts:** Personality traits predict repayment behavior



Geolocation Stability

6-12 months

15%

Tracks home/work location consistency, commute patterns, and travel frequency.



E-commerce Behavior

3+ years history

12%

Analyzes purchase patterns, return rates, payment methods, and account age.



Merchant Ratings

5+ suppliers

8%

Collects ratings from merchants/suppliers on payment reliability and business reputation.



SCORING SYSTEM

300-850 Credit Score with Explainability

Weighted model combining 6 alternative data signals into a final credit score with risk categorization and plain-language explanations.

Weight Distribution & Risk Assessment

Signal weights and risk bands applied to final calculation



Final Score: 720

Low Risk

Phone Bill (25%) 85/100

Cashflow (20%) 78/100

Psychometric (20%) 92/100

- Low (750-850)** Pre-approved
- Low-Medium (650-749)** Standard rate
- Medium (550-649)** Higher rate
- Medium-High (450-549)** Conditional
- High (300-449)** Not recommended



System Design

Multi-Agent Architecture for Credit Scoring

A modular, explainable, and bias-resilient system with 4 layers from frontend to data storage, orchestrated by 9 specialized AI agents.



Layer 1: Frontend (React)

User interfaces for applicants, bank officers, and administrators



Applicant Portal

Consent flow + psychometric questionnaire



Bank Dashboard

Score view + loan recommendation



Admin Panel

Weight config + analytics



Layer 2: Backend API (FastAPI)

Authentication, consent management, and scoring orchestration



Auth Service

JWT authentication + AES-256 encryption



Consent Engine

Per-source consent logging



Scoring API

Triggers agent pipeline



Layer 3: Multi-Agent Network (Neuro SAN)

9 specialized agents for scoring, synthesis, and explanation



Credit Coordinator

Orchestrates all scoring agents



Phone Bill Agent

Payment consistency scoring



E-commerce Agent

Purchase behavior analysis



Geolocation Agent

Location stability scoring



Psychometric Agent

Behavioral analysis



Merchant Agent

Business reputation scoring



Cashflow Agent

Bank pattern analysis



Risk Synthesizer

Combines scores → 300-850

Product Design

Three Interfaces for Complete Credit Ecosystem

Tailored experiences for applicants, bank officers, and administrators with role-specific features and value propositions.

Applicant Portal

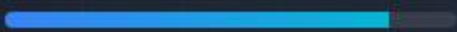
Mobile-first design

- ✓ Consent toggles for 6 data sources with privacy controls
- ✓ Psychometric questionnaire flow (10 questions)
- ✓ 60-second score result with gauge visualization
- ✓ Improvement tips and score explanation



Score: **720** Low Risk

Progress:



Trust & Transparency

Full control over data sharing with clear explanations

Bank Dashboard

Credit assessment view

- ✓ Score + risk band display with breakdown
- ✓ Per-signal visualization (6 data points)
- ✓ Loan amount & interest rate recommendation
- ✓ One-click approve/reject with audit trail



Amount: **₹2.5L** @ 12%

Risk: **Low** Score: 720

Faster Underwriting

Explainable AI decisions with full transparency

Admin Panel

System management

- ✓ Weight adjustment per segment
- ✓ Bias monitoring and outcomes tracking
- ✓ Aggregate analytics view
- ✓ Export reports and audit logs



Models: **9** Active

Accuracy: **94%** +2.3%

Governance & Control

Continuous model improvement and monitoring



Responsible AI for Credit Scoring

Bank-grade security with consent-first data handling, encryption, and explainable AI to build trust with both borrowers and lenders.



Data Protection & Security

Encryption and access control



AES-256 Encryption

All sensitive fields encrypted at rest with JWT access control for secure API interactions



Anonymized Feature Vectors

AI agents score anonymized features, never raw personal identifiers



90-Day Data Deletion

Auto-delete after 90 days unless applicant opts to retain



Consent & Transparency

User control and explainability



Consent-Based Access

Explicit YES/NO per data source with detailed logging



Explainable AI

Plain-English reasons + improvement tips for each score



Human-in-the-Loop

Bank officer makes final decision, AI provides recommendations



Build Roadmap

Build in 10 Steps, Demo with Confidence

This plan maps directly to your 'PLAN (2).md': clear implementation order, team split, and timeline to finish before presentation week.



Step 1-3: Foundation

✓ BUILT

Project setup, SQLite schema, auth + AES/JWT. Establishes secure core before agent work.



Step 4-6: Intelligence Core

✓ BUILT

Synthetic data, 9-agent Neuro SAN network, and FastAPI scoring/report routes.



Step 7-8: Product UX

✓ BUILT

Applicant wizard, score gauge, bank dashboard, and frontend-backend API integration.



Step 9-10: Test & Demo

✓ BUILT

End-to-end testing, demo seed data, and rehearsal flow for judges.



Team Split

Abhishek: Steps 1,2,3,5,6,9 (backend + agents)

Ankur: Steps 4,7,8,10 (synthetic data + frontend + demo seed)

Together: Final integration test and pitch rehearsal.



Timeline (Apr 26 - May 7)

01

Apr 26-27: Steps 1-4 (setup, DB, auth, synthetic data)

02

Apr 28-29: Step 5 (Neuro SAN agents, highest priority)

03

May 1-7: UI, integration, demo seed, fixes, PPT rehearsal

WORK BLOCK

DURATION

DELIVERABLE

OUTPUT

✓ Core Backend

~1.5 hrs

DB tables, secure auth, consent logs

Hackathon-ready backend base

Demo Script & Final Pitch

Start on Dashboard

Step 1-3

Show pre-seeded applicants, compare risk categories, and open one profile breakdown.

Run New Applicant

Step 4-7

Register, collect consents, answer 3 sample psychometric questions, submit for scoring.

Show Outcome

Step 8-9

Present gauge score, signal breakdown, recommendation, then refresh dashboard to show new entry.

What Judges See

End-to-end working system with 60-second scoring and clear explainability.

Compliance Callouts

Consent-first flow, AES-256 data protection, and human final decision checkpoint.

Prototype Honesty

Be explicit: external APIs are simulated, architecture is production-pluggable.

Closing Line

CreditBridge turns invisible borrowers into assessable, financeable customers.

Thank You — CreditBridge for Inclusive Lending

PSB Hackathon 2026 | UCO Bank × Department of Financial Services



Abhishek Kumar
LEADER



Ankur Gupta
MEMBER

 Q&A